Homework 11: 10/15 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Chain Rule

Choose 10 of the following problems to solve.

1. AC 2.5.1-14 (each of these counts as one problem)
2. APEX 2.5.7-36 EXCEPT 11, 20, 21, 22 (each of these counts as one problem)
3. PAULS-ASSIGN.Derivatives.Chain Rule.1-60 EXCEPT 14, 17, 18, 19, 20, 21, 23, 34, 39, 41, 46, 48, 53 (each of these counts as one problem)

2 Challenge Problem

In class we used the chain rule to calculate the derivative of a complicated composite function in terms of its simpler component functions. We can

also use the chain rule to work "the other direction" i.e. if we have two complicated functions whose composite is simple.

For example, take the natural logarithm function $\ln(x)$. It is defined to be the *inverse* function to e^x, so it's domain is $(0, \infty)$ and it satisfies the identities:

- $e^{\ln(x)} = x$ for all $x > 0$
- $\ln(e^x) = x$ for all real x.

Apply chain rule to the above identities to determine the derivative of $\ln(x)$.